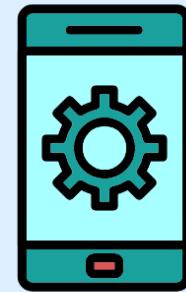


Reverse engineering

Stack, Static vs
Dynamic linking, ELF
file, PIC

Ruxandra F. Olimid

March 11, 2026



Agenda

- Approaches to RE
 - Black-box, white-box, and gray-box
- Static vs dynamic linking
- Stack
- ELF File

Approaches to RE

- Black Box vs.
White Box vs.
Gray Box

Black Box

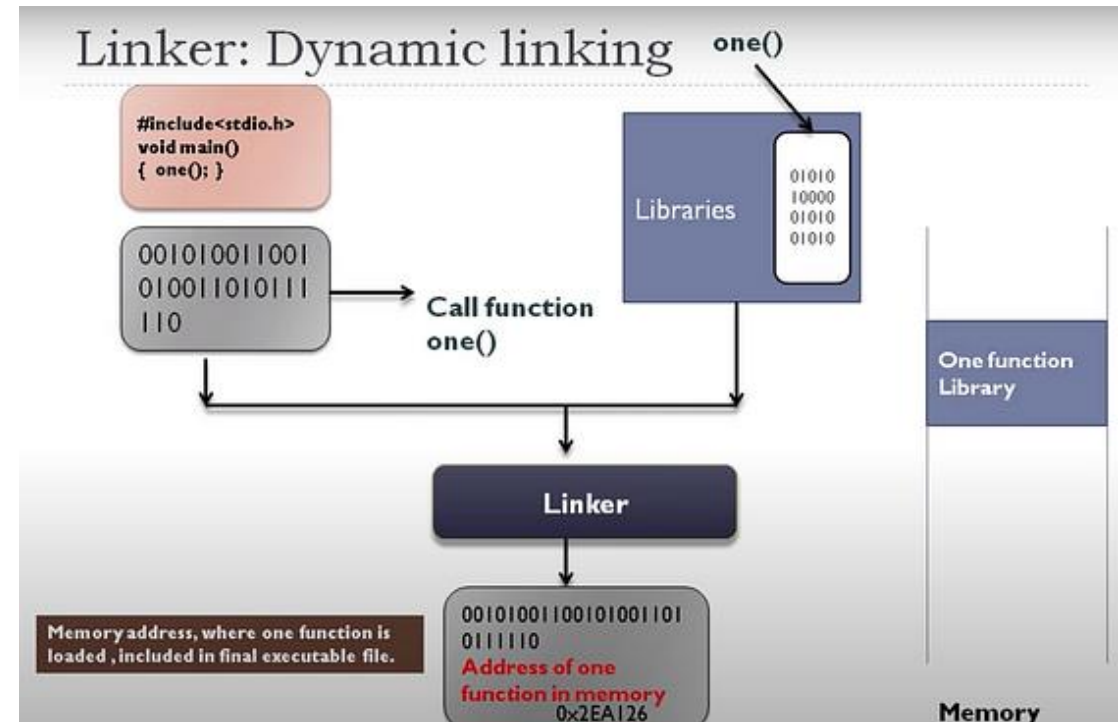
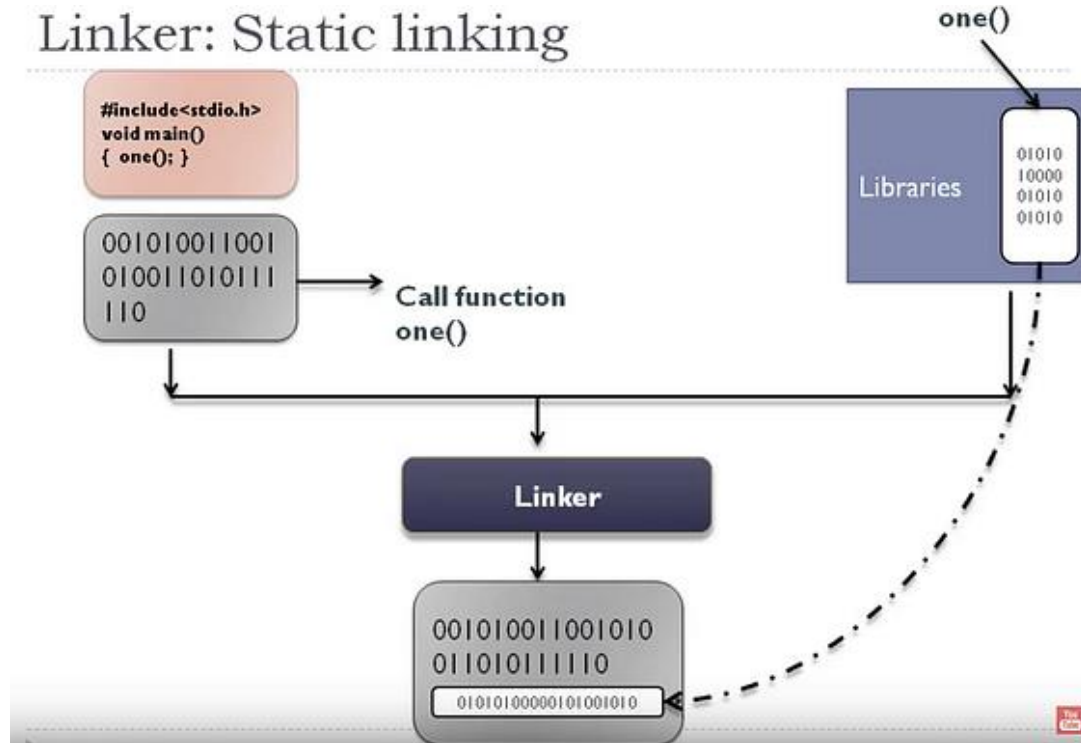
Audit software runtime
environment
External threats
Denial of service
Cascade failure
Security policy and filters
Scales and runs across
enterprise network
Valuable to security/systems
administrators

White Box

Audit software code
Programming errors
Central code repository
required
Valuable to developers and
testers

Source: Hoglund, G., & McGraw, G. (2004). Exploiting software: How to break code. Pearson Education India, online sample chapter https://www.informit.com/store/exploiting-software-how-to-break-code-9780201786958?w_ptgrevartcl=Reverse+Engineering+and+Program+Understanding_353553.

Static and dynamic linking



Source: <https://medium.com/@abhishekjainindore24/static-library-vs-dynamic-library-understanding-the-differences-26e47cac93b6>

- Read more: Medium. Static Library vs Dynamic Library: Understanding the Differences <https://medium.com/@abhishekjainindore24/static-library-vs-dynamic-library-understanding-the-differences-26e47cac93b6>

Static and dynamic linking

- Read more: Medium. Static Library vs Dynamic Library: Understanding the Differences
<https://medium.com/@abhishekjainindore24/static-library-vs-dynamic-library-understanding-the-differences-26e47cac93b6>

STATIC LINKING VERSUS DYNAMIC LINKING

STATIC LINKING	DYNAMIC LINKING
Process of copying all library modules used in the program into the final executable image	Process of loading the external shared libraries into the program and then bind those shared libraries dynamically to the program
Last step of compilation	Occurs at run time
Statistically linked files are larger in size	Dynamically linked files are smaller in size
Static linking takes constant load time	Dynamic linking takes less load time
There will be no compatibility issues with static linking	There will be compatibility issues with dynamic linking
	Visit www.PEDIAA.com

Intro to Reverse Engineering

- What to look for
 - M.Babcock , Medium. **Tracing Code Like a Hacker: Intro to Reverse Engineering**
https://medium.com/@markbabcock_79883/tracing-code-like-a-hacker-intro-to-reverse-engineering-963168f29ffc
- Instruction Pointer
 - RE Tutorial - Part 12: Instruction Pointer Register**
<https://0xinfection.xyz/reversing/pages/part-12-instruction-pointer-register.html>

Stack

- **x86 Assembly Guide**
 - **Stack during Subroutine Call**
<https://www.cs.virginia.edu/~evans/cs216/guides/x86.html>
- **The stack**
 - **RE Tutorial - Part 15: Stack**
<https://0xinfection.xyz/reversing/pages/part-15-stack.html>
- **User space vs Kernel space**
 - **RE Tutorial - Part 22 - ASM Program 1 [Moving Immediate Data]**
<https://0xinfection.xyz/reversing/pages/part-22-asm-program-1-moving-immediate-data.html>

ELF File

- Linking view (sections/linker) vs execution view (segments/loader)
 - ELF file structure
 - Program header exercise (see C.Rusu Reverse Engineering Class 0x02)
 - PIC
-
- UPB. Security Summer School **0x02. Executable File Formats**
<https://security.cs.pub.ro/summer-school/wiki/session/03>
 - Wiki OSDev <https://wiki.osdev.org/ELF>
 - Ange Albertini's (Corkami) ELF File
<https://github.com/corkami/pics/tree/master/binary/elf101>
 - Position Independent Code: <https://eli.thegreenplace.net/2011/11/03/position-independent-code-pic-in-shared-libraries>